

AI in Industry Project - CFU 3

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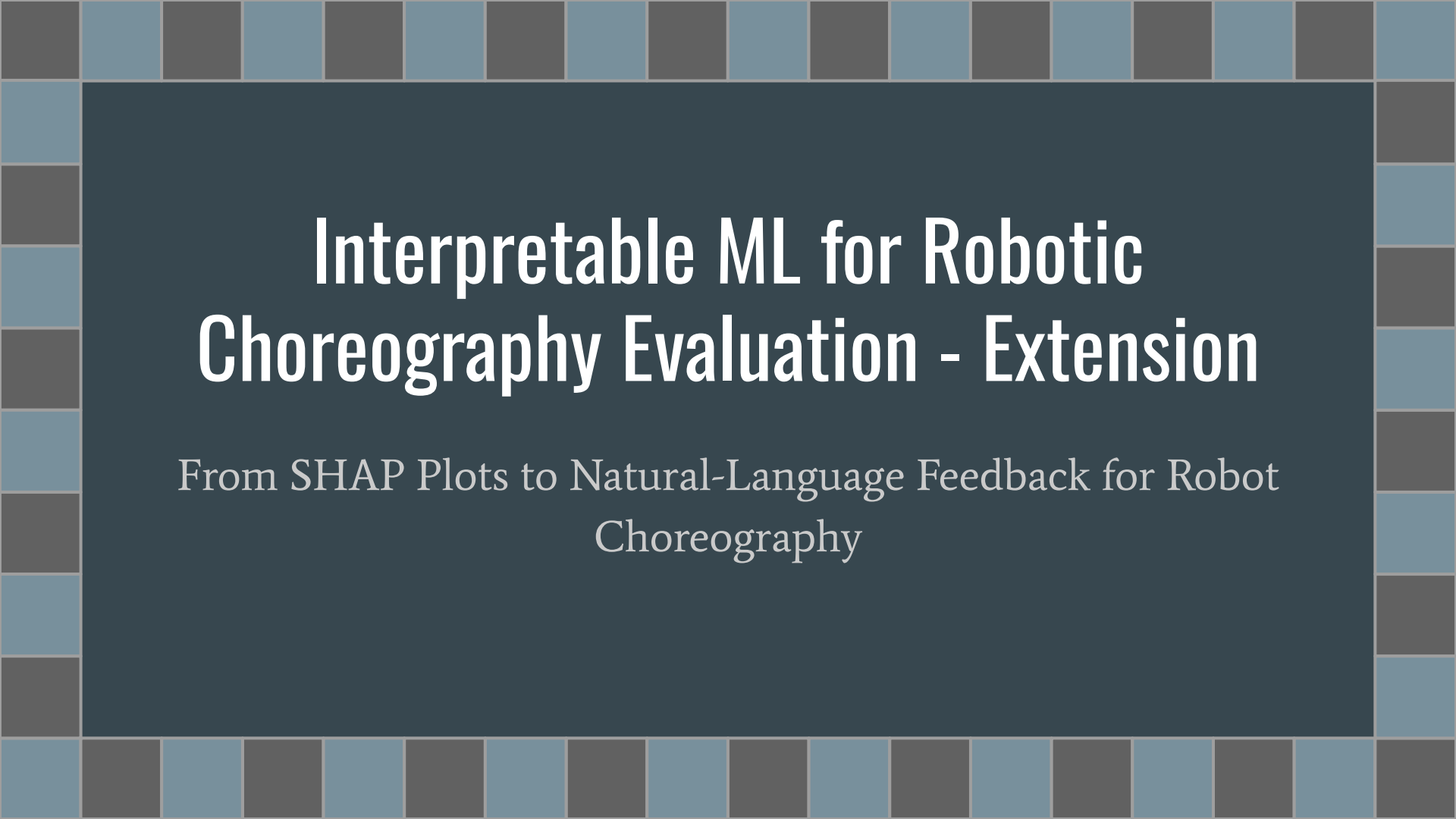
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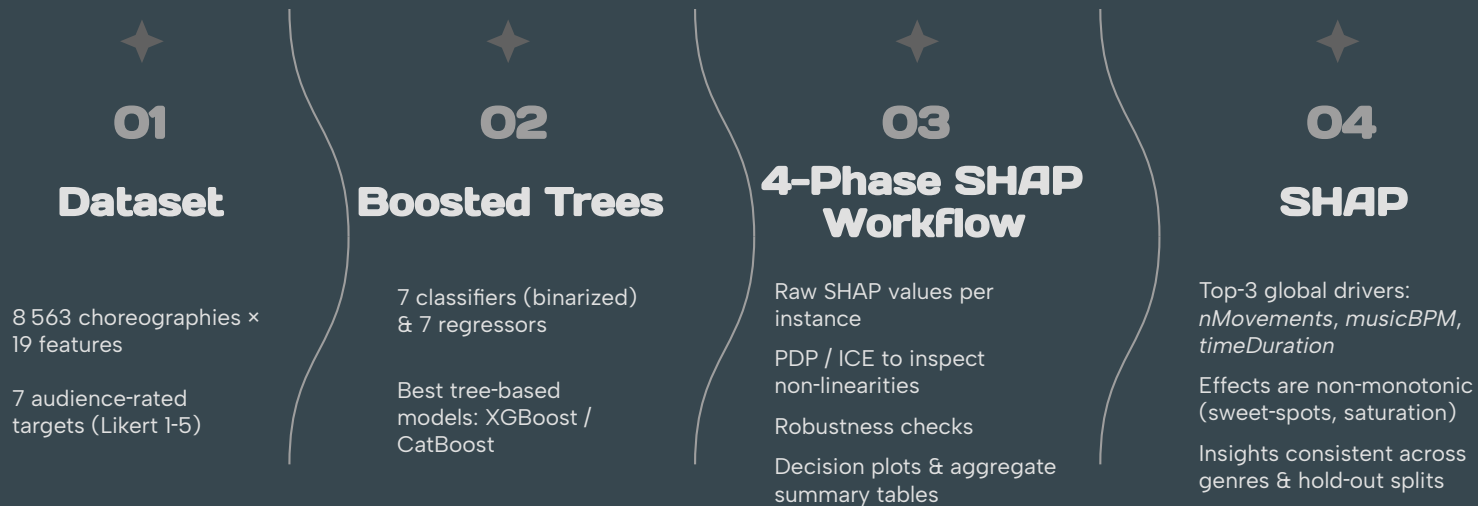


Interpretable ML for Robotic Choreography Evaluation - Extension

From SHAP Plots to Natural-Language Feedback for Robot
Choreography

Recap of Original Pipeline

- Manual evaluation $\approx$ slow & subjective
- Designers need fast, interpretable AI to support choreographers



Why an Extension?

- SHAP plots are accurate but visually heavy, so LLMs can bridge that gap quickly.
- Adds usability layer for choreographers & HRI designers
- **Goal:** Turn numeric SHAP explanations into natural-language feedback with LLMs

Extension Overview

Architecture Diagram

- Powered by Gemini 2.5-Pro LLM
- Delivered via browser UI for choreographers / HRI engineers

Input: Predicted score · SHAP vector · Raw features

LLM Prompt: Templated message

LLM Output: Feedback sentence + Improvement tip

LLM Feedback Pipeline

- **Data Ingestion:** pick 500 JSON SHAP-rich samples / target randomly
- **Prompt Engineering:** assemble system + user + few-shot template
- **LLM Generation:** Gemini returns 5-section analysis → render in UI

Prompt Design Strategy

- 5-section output:
 - Key Insights
 - Feature Analysis
 - Model Behavior
 - Practical Implications
 - Surprising Findings
- Multi-shot method
- Context awareness by briefly mention our findings

Interactive SHAP Instance Explorer UI

- Built with **Gradio 4** + **Gemini API**
- **Workflow:**
 - Choose clf/reg & target
 - Load Random Instance → features + SHAP
 - Enter API key → Ask AI

SHAP Instance Explorer

Explore SHAP explanations for robotic choreography evaluation models

Gemini API Configuration

Gemini API Key

Enter your Gemini API key here...

Configure API

API not configured

Model Type

Choose between classification or regression models

classification

regression

Target Value

Select the choreography evaluation aspect

Humancharacterization

Load Random Instance

Click 'Load Random Instance' to begin exploring!

AI Analysis

Ask AI

Load an instance first, then click "Ask AI" to get detailed analysis...

Features

Feature

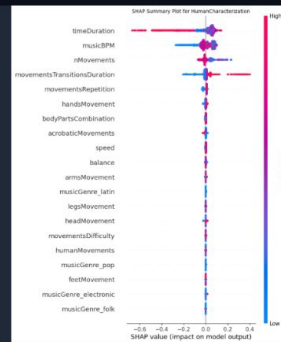
Value

SHAP Values (Ranked by Importance)

Feature

SHAP Value

SHAP Summary Plot



About SHAP Values

SHAP Instance Explorer

Instance 601 from regression model
on Space Use

SHAP Instance Explorer

SHAP Instance Explorer

Explore SHAP explanations for robotic choreography evaluation models

Gemini API Configuration

Gemini API Key

.....

Configure API ✓ Gemini API configured successfully!

Model Type

Choose between classification or regression models

☐ classification ☒ regression

Target Value

Select the choreography evaluation aspect

Spaceuse

Load Random Instance

File: instance_601.json Instance ID: 601 Prediction: 3.010145187379297

AI Analysis Task AI

Here is the analysis of the model's prediction for instance 601 on **Spaceuse**.

- Key Insights** The model predicted a Spaceuse score of 3.01, significantly driven upwards by the fast tempo of the rock music and pulled downwards by the high number of movements.
- Feature Analysis**
 - musicBPM** (Value: 147, SHAP: +0.274): This was the most significant positive contributor. The high tempo music strongly suggests the model associates faster music with more dynamic and extensive use of the available space.
 - nMovements** (Value: 20, SHAP: -0.151): This was the most significant negative contributor. A high number of movements appears to negatively impact Spaceuse, possibly because the movements become too localized and don't cover enough ground.
 - movementsTransitionsDuration** (Value: 2, SHAP: +0.013): A medium duration for movement transitions had a small positive impact, suggesting the model favors a moderate pace in transitioning between positions to effectively utilize the space.
 - timeDuration** (Value: 116, SHAP: +0.012): The duration of the choreography also had a slight positive effect, indicating that longer routines offer more opportunities for expansive space use.
 - bodyPartsCombination** (Value: 1, SHAP: -0.010): The lowest setting for body part combination had a negative effect. The model might prefer to see diverse body parts used together, so a low score here reduces the prediction.
- Model Behavior** The model seems to prioritize the overall energy and pace of the choreography (indicated by **musicBPM**) when evaluating **Spaceuse**. However, it also suggests that simply packing in a lot of movements (**nMovements**) is counterproductive. The ideal scenario, according to the model, involves a balance between a dynamic tempo and well-

Features

Feature	Value
timeDuration	116
nMovements	20
movementsDifficulty	2
robotSpeech	1
acrobaticMovements	3
movementsRepetition	2
movementsTransitionsDuration	2
humanMovements	2
balance	1
speed	2
bodyPartsCombination	1
musicBPM	147

SHAP Values (Ranked by Importance)

Feature	SHAP Value
musicBPM	0.273972
nMovements	-0.150684
movementsTransitionsDuration	0.012512
timeDuration	0.011602
bodyPartsCombination	-0.010273
movementsDifficulty	-0.009651
musicGenre_rock	0.008888
legsMovement	-0.006461
movementsRepetition	-0.005458
headMovement	0.005203
musicGenre_rap	0.002726
robotSpeech	-0.002597

SHAP Summary Plot

About SHAP Values

SHAP (Shapley Additive exPlanations) values explain the contribution of each feature to the model's prediction:

- Positive values:** Features that increase the prediction
- Negative values:** Features that decrease the prediction
- Magnitude:** Shows how much the feature contributes
- Zero:** Features with no contribution to this specific prediction

Features are ranked by absolute SHAP value from largest impact first.

Challenges

- **Prompt Sensitivity & Hallucination Risk**
 - The LLM's output quality is highly dependent on precise prompt engineering.
 - Minor changes in the prompt can lead to different or less accurate explanations.
- **Incomplete Context for LLM**
 - Providing only *local* (instance-specific) SHAP values forces the LLM to speculate about global trends and feature interactions.
 - This can lead to plausible but potentially incorrect explanations for complex model behaviors.
- **Managing the Token Budget**
 - Sending full-precision numerical data is computationally expensive.
 - We had to truncate low-impact decimals to reduce token usage while preserving key information.
- **Bridging the Language Gap**
 - A core challenge is translating technical feature names (e.g., `movementsTransitionsDuration`) into intuitive, domain-specific language that is meaningful to a choreographer.

Future Work

- **Interactive "What-If" Simulator**
 - Develop a real-time UI with sliders for choreographers to tune features (like duration or speed) and instantly see the predicted score change.
- **Multimodal Visual Explanation**
 - Integrate multimodal inputs into LLMs to directly interpret and explain SHAP summary and dependence plots, providing a richer, multi-format analysis.
- **Public Web Framework for the HRI Community**
 - Evolve the prototype into a polished, public web application with standardized metrics, creating a valuable tool for researchers and designers to benchmark their own work.

Questions?

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